

1A Linear Lithium Battery Charger Management IC with 30V Input Voltage

1 Features

- Input withstand voltage 30V
- Charge target voltage: 4.2V, support custom full voltage range: 4.05V~4.4V (step=50mV)
- Support customized lithium iron phosphate battery, charge target voltage range: 3.5V~3.8V (step=50mV)
- Maximum 1A charging current
- Optional external resistance setting charging current function or NTC function (choose one of two)
- Full stop charge detection current is 30mA
- Trickle charging current is 1/5 constant current charging current
- BAT Standby current 50nA
- Soft start to limit inrush current
- Support LED charging status indicator, support constant current function (can omit LED current limiting resistance)
- Support battery reverse protection
- Support a variety of protection functions: IC overtemperature protection, input undervoltage protection, input overvoltage protection, NTC protection (IP4054V_NTC model support)
- Package SOT23-5

2 Typical Applications

- Low power handheld devices

3 Description

IP4054V is an IC that supports linear charge management of a single lithium battery with an input withstand voltage of 30V.

IP4054V integrates power MOS, only needs very few peripheral devices, effectively reduces the size of the overall scheme and BOM cost.

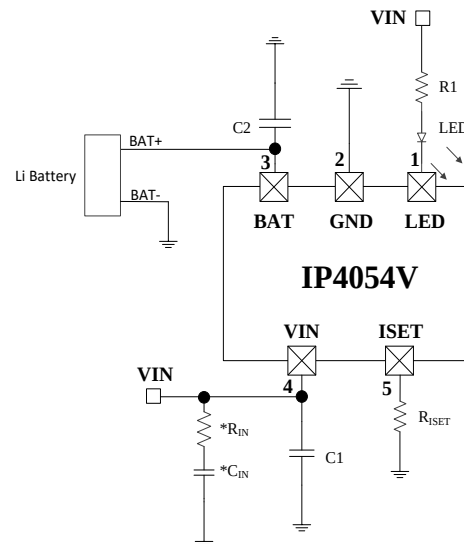
IP4054V features a complete triple charging process of Trickle Charge (TC), Constant Current (CC) and Constant Voltage (CV) charging; the Trickle Charge (TC) stage pre-charges the battery to restore a fully discharged battery; the Constant Current (CC) mode safely delivers a buck fast charge; and the final stage, Constant Voltage (CV) charging mode, ensures that the battery's full capacity is safely reached.

IP4054V has an input undervoltage protection function that intelligently adjusts the charging current and adapts to the load capacity of the adapter to prevent adapter failure.

IP4054V supports ISET pin external connection resistor to set charging current, or customized to NTC function

IP4054V is packaged in SOT23-5

4 Simplified Application Schematic



*Add *C_{IN}, *R_{IN} for more than 20V plugging.
Recommended *C_{IN}=1uF, *R_{IN}=5.1R

Figure 1 IP4054V Simplified Application Schematic

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5 Modify records

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Initial release version V1.00 (2024.3)

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6 Common Model

Type name	Function
IP4054V	The standard model, ISET pin is ISET function ,without NTC function
IP4054V_NTC	The ISET pin is NTC function,without ISET function
IP4054V_4V05	Modify the Charge Target Voltage to 4.05V from the standard model.
IP4054V_4V1	Modify the Charge Target Voltage to 4.1V from the standard model.
IP4054V_4V35	Modify the Charge Target Voltage to 4.35V from the standard model.
IP4054V_NTC_4V35_5M5	Modify the Charge Target Voltage to 4V35 and charge current to 550mA based on IP4054V_NTC
IP4054V_EN_5M	Customization of the ISET pin to EN function on standard models
IP4054V_CT	Opening of the temperature ring on the basis of the standard model
IP4054V_NTC_CT	Opening of the temperature ring on the basis of the IP4054V_NTC
IP4054V_4V35_CT	Opening of the temperature ring on the basis of the IP4054V_4V35

7 PIN Description

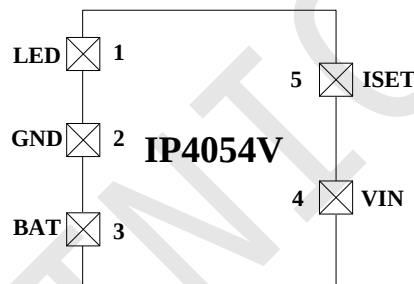


Figure 2 Pin of IP4054V

Pin No.	Pin Name	Pin Description
1	LED	Charging indicator LED, support constant current function (can omit LED current limiting resistor)
2	GND	Power Ground
3	BAT	Battery connection pin, connected to the positive electrode of the battery
4	VIN	Input pin, connect 5V charging input
5	ISET	Constant current charging setting pin, can be customized to NTC pin

8 Functional Block Diagram

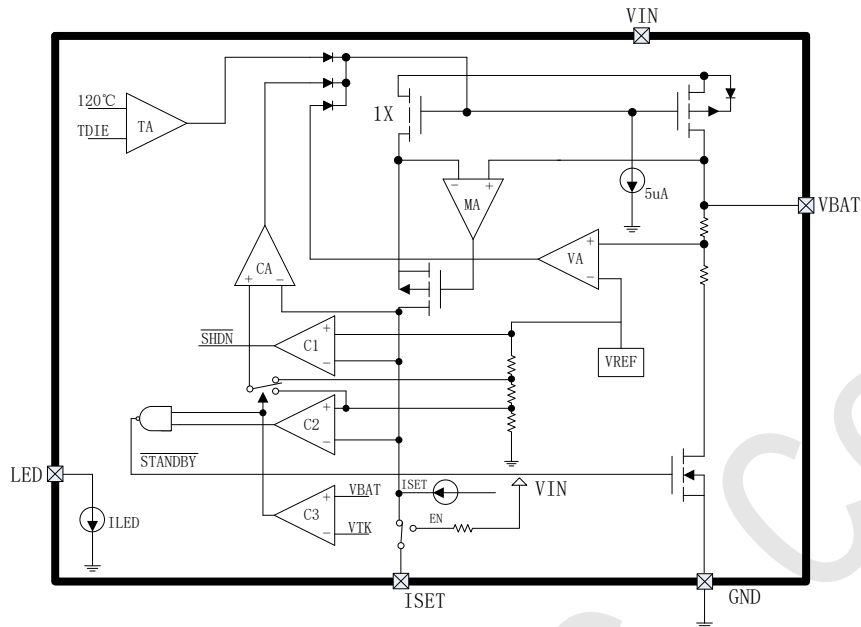


Figure 3 IP4054V Functional Block Diagram

9 Limit parameters

Parameters	Symbol	Value	Unit
Input port voltage range	V_{IN}	-0.3 ~ 30	V
Battery port voltage range	V_{bat}	-6 ~ 7.5	V
Junction temperature range	T_J	-40 ~ 150	°C
Storage temperature range	T_{stg}	-65 ~ 150	°C
Thermal resistance (junction temperature to environment)	θ_{JA}	220	°C/W
Human Body Model (HBM)	ESD	4	KV

*Stresses beyond those listed under Absolute Maximum Ratings may cause permanent damage to the device. Exposure to Absolute Maximum Rated conditions for extended periods may affect device reliability.

10 Recommended Operating Conditions

Parameter	Symbol	Min.	Typ.	Max.	Unit
Input Voltage	V_{IN}	4.5	5	5.5	V
Charge Current	I_{OUT}			1	A

* Devices' performance cannot be guaranteed when working beyond those Recommended Operating Conditions.

11 Electrical Characteristics

Unless otherwise specified, $T_A=25^{\circ}\text{C}$, $V_{IN}=5\text{V}$

Parameter	Symbol	Test Conditions	Min.	Typ.	Max.	Unit
Charging System						
Input Voltage	V_{IN}		4.5	5	5.5	V
Input under-voltage threshold	V_{IN-UV}			$V_{BAT}+0.4$		V
Input over-voltage threshold	V_{IN-OV}			6		V
Input overvoltage protection hysteresis				400		mV
Input Current	I_{VIN}	$V_{IN}=5\text{V}$, $V_{BAT}=\text{NC}$, NO LED		1	2	mA
Standby Current	$I_{\text{standby-BAT}}$	$V_{IN}=0\text{V}$, $V_{BAT}=3.7\text{V}$, NO LED		50	2000	nA
		$V_{IN}=0\text{V}$, $V_{BAT}=-6\text{V}$, NO LED		1	2	mA
Charge Current	I_{CC}	$R_{ISET}=1\text{K}$	0.9	1	1.1	A
		$R_{ISET}=\text{NC}$ ($\geq 120\text{K}$)	0.45	0.5	0.55	A
Charge Target Voltage	V_{CV}	$V_{IN}=5\text{V}$	4.16	4.2	4.24	V
Full charge stop detection voltage	V_{SV}			$V_{CV}-0.05$		V
Charging voltage after full charge	V_{RC}			$V_{CV}-0.1$		V
Trickle over constant current voltage	V_{TK}	$V_{IN}=5\text{V}$	2.9	3.0	3.1	V
Trickle Charge Current	I_{TK}	$V_{IN}=5\text{V}$, $V_{BAT}<2.9\text{V}$		$1/5 I_{CC}$		mA
Charge Cut-off Current	I_{STOP}			30	50	mA
Control System						
LED drive Current	I_{Led}	$V_{IN}=5\text{V}$			5	mA
Thermal shutdown temperature	T_{OTP}	Rising Threshold	100	110	120	$^{\circ}\text{C}$
Thermal shutdown recovery temperature	T_{OTP-H}	Falling Threshold	80	90	100	$^{\circ}\text{C}$

12 Function Description

12.1 Charge Process

The IP4054V uses a full trickle/constant/constant voltage charging mode.

When the battery voltage is less than the trickle to constant current voltage V_{TK} , it is charged with trickle charging current I_{TK} .

When the battery voltage is greater than V_{TK} , charge with constant current charging current I_{CC} .

When the battery voltage approaches the set constant voltage charging voltage V_{CV} , the charging voltage V_{CV} remains unchanged, the charging current slowly decreases, and the constant voltage charging mode is entered.

After entering the constant voltage charging mode, if the charging current is less than the full charge stop detection current I_{STOP} . The charging will be stopped first, and then detect whether the battery voltage is higher than the stop voltage V_{SV} . If it is higher than the charging stop voltage V_{SV} , stop charging. If the stop voltage is lower, charging continues.

After the battery is fully charged and stopped, and the input V_{IN} continues to be active, if the battery voltage is less than V_{RC} , it will enter the full charge stage and start the charging process again.

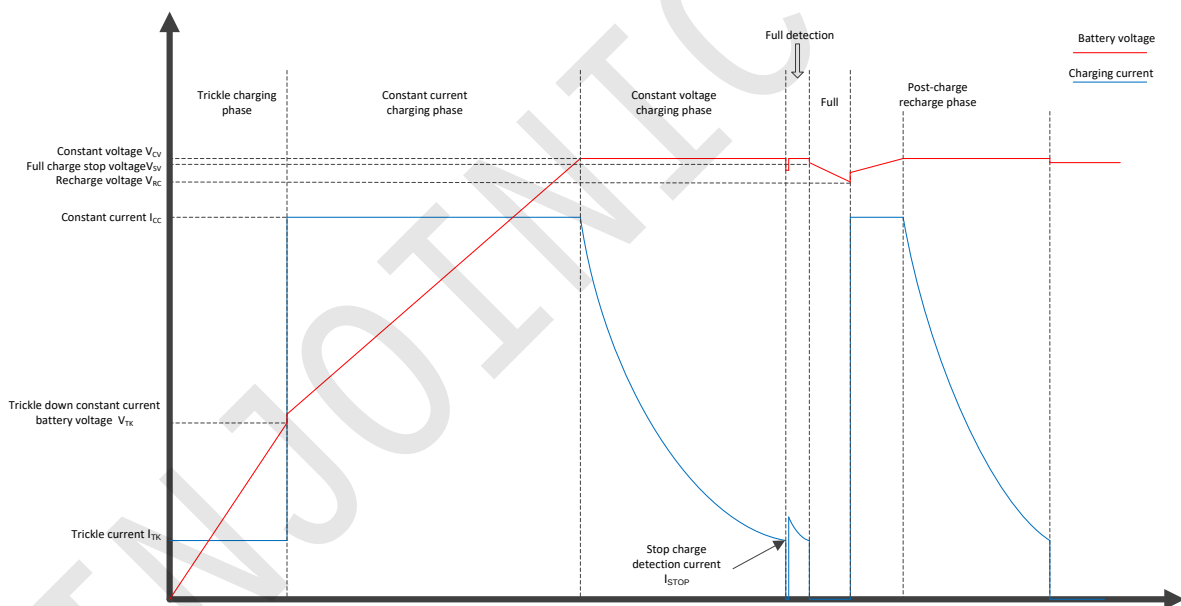


Figure 4 IP4054V Schematic diagram of the charging process

12.2 Charging protection

IP4054V has perfect protection function, integrated input under voltage, input over voltage, IC over temperature protection and other functions to ensure stable and reliable work of the system.

The IP4054V has an input V_{IN} withstanding voltage of 30V and will stop charging if the input V_{IN} is greater

than 6V.

The IP4054V features VIN input undervoltage protection, which automatically adjusts to reduce the charging current when it detects that the input voltage is close to $V_{BAT}+0.4V$, ensuring that it does not pull the adapter.

IP4054V integrated over-temperature protection function, when the internal temperature of the chip is detected to exceed 110 degrees, it will be forced to stop charging.

12.3 Charge current setting

The IP4054V standard product supports this function

IP4054V supports ISET pin to set the charging current, external resistor R_{ISET} to set the constant charging current;

The relationship between constant current and R_{ISET} : $I_{CC}=1000/R_{ISET}$.

R _{ISET} resistance (Ω)	constant charging current
1K	1A
2K	0.5A
10K	0.1A
NC (≥120K)	0.5A

12.4 NTC

The IP4054V_NTC model supports the NTC function (the IP4054V_NTC does not support the ISET pin to set the charge current function)

IP4054V_NTC supports NTC protection function which can cooperate with NTC resistance to detect battery temperature;

By default, 100uA current is discharged, and when the pin voltage is detected to be greater than 1.5V (NTC resistance is greater than 15K), the output current is reduced to 30uA; at the output of 30uA, when the pin voltage is detected to be less than 0.3V (NTC resistance is less than 10K), the output current changes to 100uA.

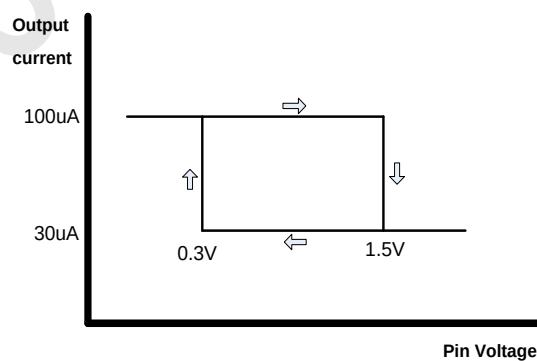


Figure 5 Pin output current vs. pin voltage

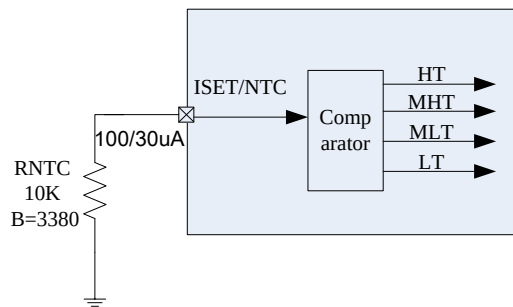


Figure 6 NTC Diagram

When IP4054V detects that the NTC pin voltage rises to greater than $0.82V@30\mu A$ (below $0^{\circ}C$), it triggers low-temperature protection and stops charging;

When IP4054V detects that the NTC pin voltage is between $0.54V@30\mu A \sim 0.82V@30\mu A$ ($0^{\circ}C \sim 10^{\circ}C$), it triggers the medium-low temperature protection and the charging current is reduced to half;

When IP4054V detects NTC pin voltage between $0.49V@100\mu A \sim 0.54V@30\mu A$ ($10^{\circ}C \sim 45^{\circ}C$), it indicates that the battery temperature is normal and normal charging;

When IP4054V detects NTC pin voltage between $0.417V@100\mu A \sim 0.49V@100\mu A$ ($45^{\circ}C \sim 50^{\circ}C$), it triggers medium-high temperature protection with full voltage $CV-50mV$;

When IP4054V detects that the NTC pin voltage drops to less than $0.417V@100\mu A$ (above $50^{\circ}C$), it triggers high temperature protection and stops charging.

If NTC function is not required, connect the NTC pin to ground with a 10K resistor.

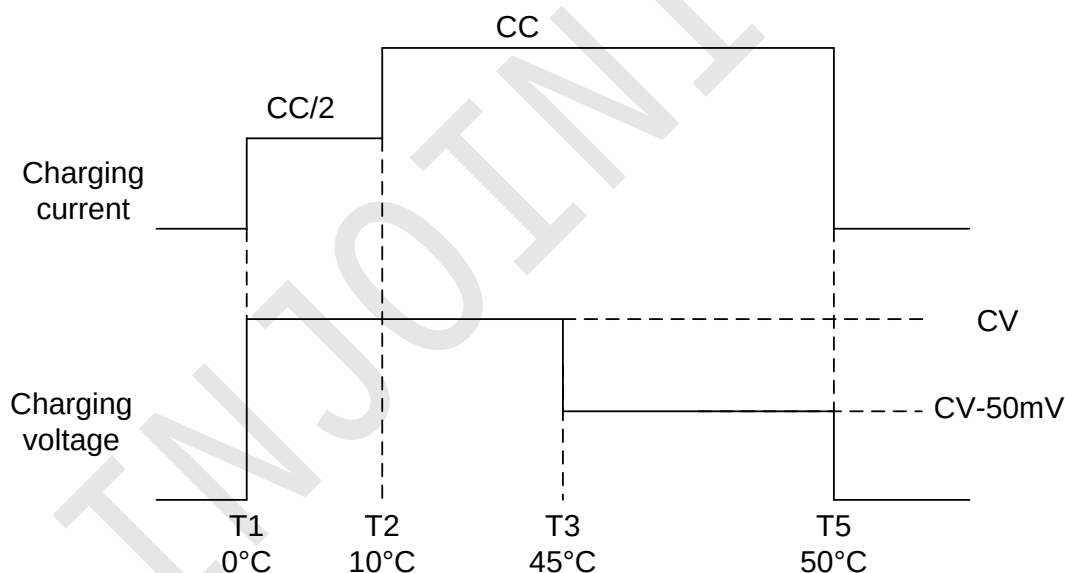


Figure 7 Schematic diagram of 5-segment battery temperature charge protection (JEITA compliant)

12.5 Charging LED indication

Standard product light display for: charging process LED on, charging full LED off, abnormal state (over-temperature protection, NTC protection) LED flashes (0.5S on, 0.5S off).

12.6 Battery reverse protection

IP4054V supports reverse battery protection, which can effectively prevent the chip from being damaged when the battery is reversed.

When $V_{IN}=0V$, the maximum negative withstand voltage at the BAT terminal is $-6V$.

When $V_{IN}=5V$, the maximum negative withstand voltage of BAT is $-5V$.

13 Typical Application Schematic

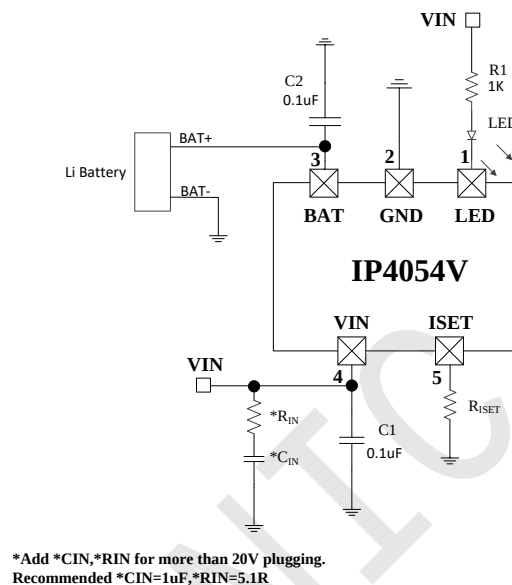


Figure 8 Typical Application Schematic

14 BOM

No.	Part Name	Type & Specification	Units	Quantity	Location	Note
1	IC	IP4054V	PCS	1	U1	
2	SMD capacitors	0603 0.1uF 25V 10%	PCS	2	C ₁ 、C ₂	
3	SMD resistors	0603 1K 5%	PCS	1	R1	
4	SMD resistors	0603	PCS	1	R _{ISET}	
5	LED	0603	PCS	1	LED	
6	SMD capacitors	0603 1uF 50V 10%	PCS	1	*CIN	Add when plugging or unplugging above 20V
7	SMD resistors	0603 5.1R 5%	PCS	1	*RIN	

15 Silkscreen

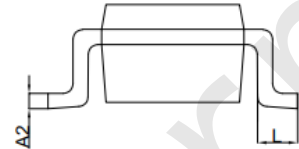
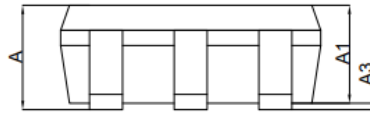
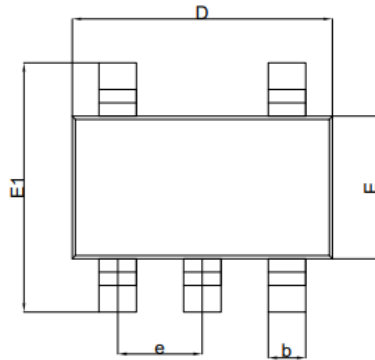


Instruction:

- 1、4054V --Product name
- 2、XXXXX --Product number
- 3、○ --PIN1 Position

Figure 9 IP4054V Silkscreen

16 Package



	POD	SOT23-5		
		Size unit: mm		
	Symbol	Minimum	Normal	Maximum
Total Thickness	A	-	-	1.25
Molding Thickness	A1	1.00	1.10	1.20
LF Thickness	A2	0.15	-	0.19
Stand Off	A3	0.04	-	0.10
Body Size	D	2.82	2.92	3.02
	E	1.50	1.60	1.70
	E1	2.60	2.80	3.00
Lead Width	b	0.33	-	0.41
Lead Length	L	0.30	-	0.60
Lead Pitch	e	0.95 BSC		

SOT23-5

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